ROADMAP: FUNDED PHD (FALLBACK MSC), START FALL 2028

Thread: reliable ML for noisy real-world signals **Time:** ~11 hrs/wk (1h/day + 2h Fri/Sat)

PHASE 1 | Nov–Dec 2026 — FOUNDATIONS

- Update portfolio: papers out + one-line thread statement
- Math pass (1-2 wks): linear algebra, vector calculus basics
- Karpathy micrograd: backprop by hand, once
- PyTorch official tutorials; write your own training loop
- Weekly check: explain the code aloud with no AI

> **EXIT:**
> A 1D CNN on PTB-XL that you wrote and can explain

PHASE 2 | Jan–Jun 2027 — PROOF

- Thesis experiments (teacher/student, noise tests, ECE)
- Rebuild your demodulator pipeline in PyTorch unaided; keep a "stuck list"
- Get SDR access via MIST
- Clean GitHub: README + runnable code
- Basic stats: hypothesis tests, calibration

> **EXIT:**
> You can defend every paper and the thesis with zero AI

PHASE 3 | Jun–Sep 2027 — TARGETING

- PI sheet: 15 names (5 reach / 5 match / 5 safe)
  - columns: lab, 2 recent papers, one honest connection
- Draft research statement around ONE identity
- Book IELTS/TOEFL; GRE only if a target requires it
- Ask Dr. Grgic + one BUP professor for letters early

> **EXIT:**
> Shortlist locked

PHASE 4 | Oct 2027–Jan 2028 — APPLY

- Portal applications first (check each deadline)
- PI emails: (1) their paper -> your link, (2) a specific contribution, (3) follow-up with new value after 2 weeks
- Interview prep: defend your work cold

> **EXIT:**
> Ready for admission & funding interviews

RULE

If you can't explain it, you can't cite it.